PES UNIVERSITY

LAB 3: COMPONENT MODELLING & ARCHITECTURAL & PATTERN SELECTION

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Problem Statement : ***Multi-Cloud Asset Cost Optimization Portal***

The system helps **FinOps analysts and cloud administrators manage and optimize cloud costs across AWS, Azure, and GCP** by ingesting billing data, identifying idle or unused cloud assets, estimating potential cost savings, and dispatching budget breach alerts.

REQUIREMENTS TABLE :

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall ingest cloud billing CSV exports from **AWS, Azure, and GCP**, categorize costs by service and project tag, and highlight compute instances with **less than 5% CPU utilization over 14 days**	High	**Pass:** An idle cloud asset is flagged and a monthly savings estimate is displayed. **Fail:** An active instance is incorrectly classified as idle.	Helps the FinOps Analyst identify unnecessary cloud spending and potential cost savings across multiple cloud providers.
FR-002	Functional	The system shall identify **unattached storage volumes and idle compute instances** across AWS, Azure, and GCP.	High	**Pass:** Unattached storage volumes and idle compute instances are correctly identified and displayed. **Fail:** An active or attached cloud asset is incorrectly flagged as idle or unused.	Helps the FinOps Analyst and Cloud Administrator find unused cloud resources that may be increasing costs unnecessarily.
FR-003	Functional	The system shall calculate and display the **estimated monthly cost savings** for identified idle or unused cloud assets.	High	**Pass:** A monthly savings estimate is displayed for each identified cost-optimization opportunity. **Fail:** An identified idle asset has no savings estimate.	Allows the FinOps Analyst to understand the potential financial benefit of removing or optimizing unused cloud resources.
FR-004	Functional	The system shall provide **cost analytics and dashboards** that present cloud spending information categorized by service and project tag.	High	**Pass:** The dashboard displays cloud costs categorized by service and project tag. **Fail:** Costs are missing or cannot be categorized correctly.	Enables the FinOps Analyst to analyze cloud spending and identify areas where costs can be optimized.
FR-005	Functional	The system shall **dispatch alerts**	High	**Pass:** An alert is dispatched when spending breaches	Helps the FinOps Analyst

		when cloud spending exceeds configured budget thresholds.		the configured budget threshold. **Fail:** No alert is dispatched after a budget breach.	and Cloud Administrator respond quickly to unexpected or excessive cloud spending.
NFR-001	Performance & Security **NON-FUNC**	The portal shall parse **multi-gigabyte cloud billing files** and update cost dashboards in **under 2 minutes**, while maintaining required security standards	High	**Pass:** Benchmarking tests confirm the dashboard update target and security standards under simulated peak load. **Fail:** The portal exceeds the 2-minute target or fails the required security checks.	Ensures the portal can efficiently process large billing datasets while protecting cloud cost information
NFR-02	Security **NON-FUNC**	The portal shall ensure that **cloud billing data and cost information are protected from unauthorized access**	High	**Pass:** Only authorized users can access billing data and cost dashboards. **Fail:** An unauthorized user can access protected cloud cost information.	Protects sensitive financial and cloud infrastructure information used by FinOps Analysts and Cloud Administrators.

1. Architectural Style Analysis

Compare:

- Layered Architecture
- Microservices Architecture
- Client-Server Architecture

Architectural Style	Advantages	Disadvantages	Suitability for Multi-Cloud Asset Cost Optimization Portal
Layered Architecture	Clear separation of concerns, easy to understand, suitable for traditional applications.	Performance overhead, tight coupling between layers, difficult to scale individual components.	**Moderate** – simple to design and maintain, but scaling different cloud-analysis functions independently can be difficult.
Microservices Architecture	Independent scaling, fault isolation, technology diversity, and independent deployment.	Operational complexity, network latency, and data consistency challenges.	**High** – suitable because the portal handles multiple cloud providers and different functions such as billing ingestion, cost analysis, asset optimization, and alerts.
Client-Server Architecture	Centralized control, simple deployment, and easy data consistency.	Single point of failure, scalability bottleneck, and limited fault tolerance.	**Low** – centralized architecture may become a bottleneck as cloud billing data and analysis workloads increase.

Conclusion

Microservices Architecture is the most suitable architecture for the Multi-Cloud Asset Cost Optimization Portal because its services can be developed, deployed, and scaled independently. This is particularly useful for handling billing

data from AWS, Azure, and GCP and for separating functions such as cost analysis, asset optimization, and budget alerts.

2. Architecture Selection

Selected Architecture: Microservices Architecture

We chose **Microservices Architecture** for the Multi-Cloud Asset Cost Optimization Portal.

Reason 1: The system handles cloud billing data from **AWS, Azure, and GCP**, so separate services can manage different cloud-provider data independently.

Reason 2: Functions such as billing ingestion, cost analysis, idle asset identification, savings calculation, and budget alerts can be developed and scaled independently.

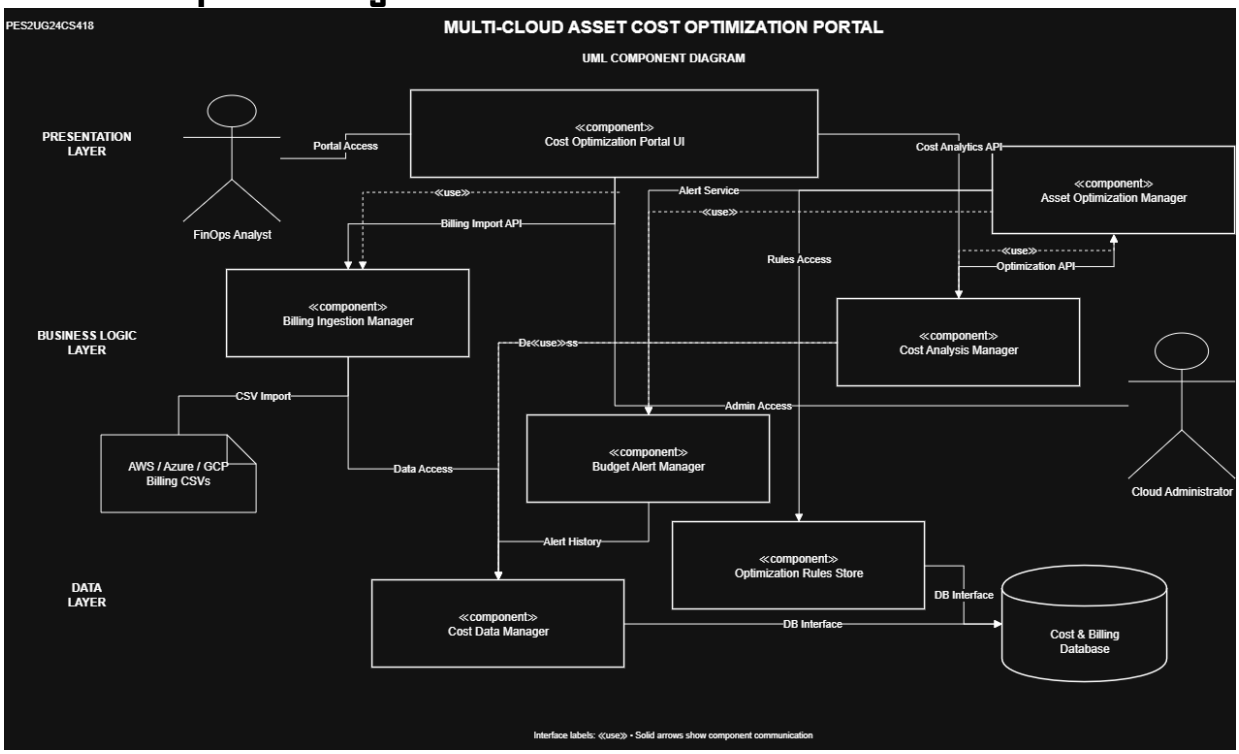
Security Advantage: Each service can apply controlled access and authentication to protect sensitive cloud billing and cost information.

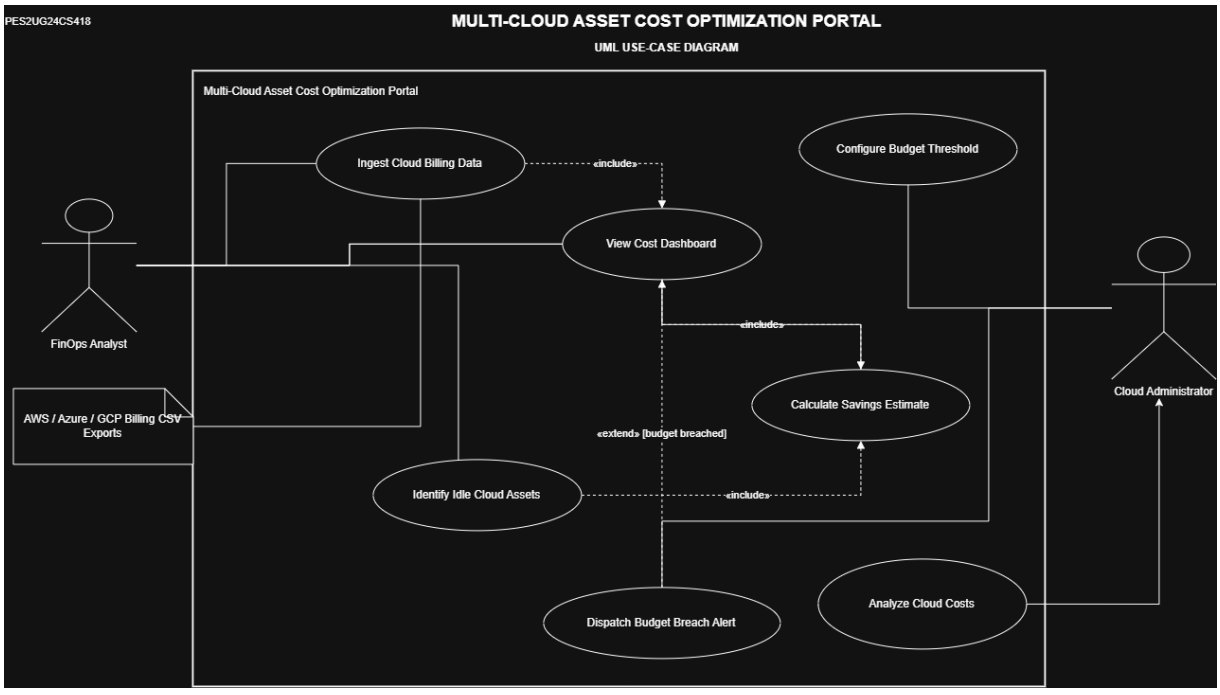
Performance Benefit: Individual services can be scaled independently according to workload, improving the portal's response when large billing files are processed.

3. Components Identified

1. **Billing Ingestion Component**
2. **Cost Analysis Component**
3. **Asset Optimization Component**
4. **Budget Alert Component**
5. **Cost Data Management Component**
6. **Cloud Provider Integration Component**
7.

4. UML Component Diagram





Presentation Layer

- Cost Optimization Portal UI

Business Logic Layer

- Billing Ingestion Manager
- Cost Analysis Manager
- Asset Optimization Manager
- Budget Alert Manager

Data Layer

- Cost Data Manager
- Optimization Rules Store
- Cost & Billing Database

5.Justification

Architecture Selection

We chose **Microservices Architecture** for the **Multi-Cloud Asset Cost Optimization Portal**.

Architectural Choice

Microservices Architecture is suitable for the portal because the system performs multiple independent functions such as billing data ingestion, cloud cost analysis, asset optimization, and budget alert management.

Reason 1

The portal processes billing information from **AWS, Azure, and GCP**. Using separate services allows each cloud-provider integration and billing-processing function to operate independently.

Reason 2

The major functions of the portal, such as **cost analysis, idle asset identification, savings estimation, and budget alerts**, can be developed and scaled independently according to their workload.

Security Advantage

Microservices allow security controls and access permissions to be applied separately to individual services. This helps protect sensitive cloud billing and cost information from unauthorized access.

Performance Benefit

Individual services can be scaled independently when the workload increases. This helps the portal process large billing datasets and provide cost information efficiently.